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## Chapter I. The Real Number System.

$\mathbb{R}$ : the set of real numbers.

Do we really know what this set is? Most objects in math are artificial. it's all about coming up with a good defn. (consistent and useful)

⚠ Depending on your taste and past experience. this may be the most difficult chapter for you.

key: abandon all the common knowledge you knew about $\mathbb{R}$. approach everything from only axioms and defns.

☺ This chapter, however, is a great opportunity to sharpen your proof writing skills. Since we know so little. the proof ideas are usually quite straightforward to identify.

*. Start from the very beginning:

① Natural numbers ($\mathbb{N}$): $\{1, 2, 3, 4, \cdots\}$.
they do have physical meanings. but one cannot perform subtraction: e.g. $3-4 \notin \mathbb{N}$.

② Integers ($\mathbb{Z}$): $\{\cdots, -3, -2, -1, 0, 1, 2, 3, \cdots\}$
things are already artificial. $+, -, \times$ are fine. but one cannot perform division: e.g. $\frac{-2}{3} \notin \mathbb{Z}$.

③ Rational numbers ($\mathbb{Q}$): $\{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\}$.
$+ - \times \div$ are all fine! And this is in fact a

very good artificial object: you can compare two rational numbers! ("ordered set") This coincides with real life needs and makes $\mathbb{Q}$ still a "one-dimensional" object.

☹ Unfortunate drawback: $\mathbb{Q}$ has holes.




Rope. want to assign a rational number to each point on it. This is impossible. Informally, if you take a sequence of rational numbers, they may eventually approach something that is not in $\mathbb{Q}$. (i.e. $\mathbb{Q}$ is NOT complete under the limit operation).

Goal: Further enlarge $\mathbb{Q}$ to a set that is still

(1) ordered (2) can perform $+ - \times \div$ ("field")

(2) Complete under Limit.

We'll use this as exactly the defn of real numbers $\mathbb{R}$. Before we introduce the formal defn. Let's look at another reason why we need to enlarge $\mathbb{Q}$ — Solve equations.

Ex! $x^2 = 2$ cannot be solved by any rational number.

Pf. Suppose this is possible. then $\exists \frac{p}{q} \in \mathbb{Q}$ s.t. $(\frac{p}{q})^2 = 2$. Obviously. $q \neq 0$. and we can choose $p, q$ s.t. $(p, q) = 1$. Since $\frac{p^2}{q^2} = 2$. one has $p^2 = 2q^2$. Hence. 2 is a factor of $p^2$. $\Rightarrow$ $p$ is even.

Assume then $p = 2k$, for some $k \in \mathbb{Z}$. $\Rightarrow$

$(2k)^2 = 2q^2 \Rightarrow 2k^2 = q^2 \Rightarrow q$ is even too!

This contradicts our assumption $(p, q) = 1$. $\square$

Exercise: Let $p$ be a prime number. prove that $x^2 = p$ cannot be solved by any rational number.

- We'll show later that $x^2 = 2$ is solvable in $\mathbb{R}$. and we write its solution as $\sqrt{2}$ (or $-\sqrt{2}$). This is a real number (needs proof) and is irrational. (Rudin, Thm 1.21)

Defn. The real numbers, denoted as $\mathbb{R}$, is a set so that

(i) it is an ordered field.

(ii) it contains $\mathbb{Q}$ as a sub-field, and $\mathbb{Q}$ obeys the same order.

(iii) it satisfies the Completeness Axiom (also known as the "least upper bound property")

Rmk. This is a good defn. since such a field exists (see Rudin P17. Appendix, or the note on "Dedekind cuts" for a construction) and is unique (in the sense of isomorphism).

## §1.1 Ordered Field

Defn. A set $F$, equipped with two operations ("addition" and "multiplication"), is called a field, if it satisfies the following "field axioms" (A), (M), (D):

(A) Axioms for addition:

(A1) $x \in F$, $y \in F \Rightarrow x + y \in F$ (sum is uniquely defined)

(A2). (Commutative) $x+y = y+x, \quad \forall x, y \in F.$

(A3) (associative) $(x+y)+z = x+(y+z). \quad \forall x, y, z \in F.$

(A4). There is an element $0 \in F$ s.t. $0+x = x. \ \forall x \in F.$

(A5). $\forall x \in F$, $\exists$ an element $-x \in F$ s.t. $x+(-x) = 0.$

(M). Axioms for multiplication:

(M1) $x \in F, \ y \in F \Rightarrow xy \in F.$ (product is uniquely defined)

(M2) (commutative) $xy = yx, \quad \forall x, y \in F$

(M3) (associative) $(xy)z = x(yz). \quad \forall x, y, z \in F.$

(M4). There is an element $1 \neq 0$ in $F$ s.t. $1x = x. \ \forall x \in F.$

(M5). $\forall x \in F$ s.t. $x \neq 0$. $\exists$ an element $\frac{1}{x} \in F$ s.t. $x \cdot \frac{1}{x} = 1.$

(D) The distributive law: $x(y+z) = xy + xz, \ \forall x, y, z \in F.$

Ex1 Prove that elements 0, 1 are unique.

Pf. (We only prove for "0", "1" can be argued similarly).
Suppose there are two elements 0 and 0' in F, both satisfying (A4). then

$$0 \underset{(A4)}{=} 0+0' \underset{(A2)}{=} 0'+0 \underset{(A4)}{=} 0'. \qquad \square$$

Ex3. Prove that $\forall x \in F$, $-x$ is unique. Similarly, if $x \neq 0$, then $\frac{1}{x}$ is also unique.

Pf. (We only look at "$-x$") Suppose for $x \in F$, there are two qualifying $y, y' \in F$ (as "$-x$"). then

$$x+y = 0, \quad x+y' = 0 \qquad (A5)$$

$$x+y=0 \Rightarrow x+y+y' = 0+y' \overset{(A4)}{=} y'$$

On the other hand, $x+y+y' \overset{(A2)}{=} x+y'+y \overset{(A5)}{=} 0+y \overset{(A4)}{=} y$

hence $y'=y$. i.e. Such $y$ is unique. □

Rmk. Exactly because of the uniqueness, we are able to introduce the notation "$-x$", "$\frac{1}{x}$". These also naturally define Subtraction and division on $F$:

"$x-y := x+(-y)$", "$\frac{x}{y} := x\left(\frac{1}{y}\right)$" (if $y\neq 0$)

Prop! Let $x, y, z \in F$. a field. then

(A). If $x+y = x+z$. then $y=z$.

(B). $-(-x) = x$ (C) If $xy = xz$. and $x\neq 0$. then $y=z$.

(D). $\frac{1}{\frac{1}{x}} = x$. (if $x\neq 0$). (E). $0x = 0$ (F) $-0 = 0$.

(G). $(-1)\cdot x = -x$ (H) $xy = 0$ iff $x=0$ or $y=0$.

(I). $(-x)(-y) = xy$.

(Left as HW or exercises. Many of the proofs can be found in Rudin or other books.)

Some sample proofs.

(A). By (A5). there exists $-x \in F$ and

$$x+y = x+z \Rightarrow x+y+(-x) = x+z+(-x)$$

$$\overset{(A3)}{\Rightarrow} (x+(-x))+y = (x+(-x))+z$$

$$\overset{(A5)}{\Rightarrow} 0+y = 0+z \overset{(A4)}{\Rightarrow} y = z.$$

(F). We have already proved the uniqueness of the element $0\in F$ satisfying (A4). it suffices to show that $-0$

also satisfies (A4). To see this. $\forall x \in F$.

$$-0+x \overset{(A4)}{=} -0+(0+x) \overset{(A3)}{=} (-0+0)+x \overset{(A5)}{=} 0+x \overset{(A4)}{=} x$$

Hence, $-0$ satisfies (A4) and $-0=0$ by uniqueness. □

**Defn.** A field $F$ is called an **ordered field** if it can be equipped with a relation "$<$" (called an "order") so that

(O1) $\forall x, y \in F$. exactly one of $x<y$, $x=y$, $y<x$ holds.

(O2) $\forall x, y, z \in F$. if $x<y$ and $y<z$. then $x<z$.

(O3). $\forall x, y, z \in F$. if $x<y$. then $x+z<y+z$.

(O4). $\forall x, y \in F$. if $0<x$ and $0<y$. then $0<xy$.

**Rmk.** ① A set satisfying (O1). (O2) is often called an ordered set. Ordered field is basically a field + an ordered set + the order works well with field operations.

② "$x>y$" is defined as "$y<x$"; "$x \leq y$" is defined as "$x<y$ or $x=y$". Similarly for "$\geq$".

③ We say $x \in F$ is positive if $x>0$, negative if $x<0$.

**Prop2.** Let $F$ be an ordered field and $x, y, z, w \in F$. Then

(A) $x>y$ and $z>0 \Rightarrow xz>yz$

(B). $x>y$ and $z<0 \Rightarrow xz<yz$

(Exercise)

(C). $x>y$ iff $-x<-y$.

(D). $x<y$ and $z<w \Rightarrow x+z<y+w$

(E). $0<x<y \Rightarrow 0<\frac{1}{y}<\frac{1}{x}$

(F). $\forall x \neq 0$. $x^2>0$. In particular. $1>0$.

Some sample proofs.

(A) Adding $-y$ to "$x>y$". by (O3). one has $x-y>y-y=0$.

Since both $x-y>0$ and $z>0$. by (O4). one has

$(x-y)\cdot z>0$. By distributive law.

$(x-y)\cdot z = xz+(-y)\cdot z = xz-yz$. So $xz-yz>0$.

Adding $yz$ on both sides. by (O3).

$(xz-yz)+yz > 0+yz = yz$.

$LHS = xz+(-1)\cdot yz+yz \overset{(D)}{=} xz+[(-1)+1]\cdot yz = xz+0\cdot yz$

$= xz+0 = xz$. Hence. $xz>yz$.

(B) Since $z<0$. from (O3). $z+(-z)<0+(-z)=-z$.

So $0<-z=(-1)\cdot z$. Thus.

$x>y \overset{(A)}{\Rightarrow} x\cdot(-1)\cdot z > y\cdot(-1)\cdot z$

$\Rightarrow -(xz) > -(yz)$.

Adding $xz+yz$ on both sides and applying (O3).

$(xz+yz)-(xz) > (xz+yz)-(yz)$

$\Rightarrow yz>xz \Rightarrow xz<yz$. □

- We now understand the defn of $\mathbb{R}$ except for the Completeness Axiom (CA). In fact. under the usual addition and multiplication. $\mathbb{Q}$ is already an ordered field. (but it doesn't satisfy CA). Our main point is that there exists a larger ordered field ($\mathbb{R}$) containing $\mathbb{Q}$, where CA holds true. and where the addition, multiplication, order are consistent with

those on $\mathbb{Q}$. (i.e. if $x, y \in \mathbb{Q}$ then $x+y$, $xy$ are defined in $\mathbb{R}$ the same way they are on $\mathbb{Q}$. and in particular $x+y, xy \in \mathbb{Q}$. Also, if $x<y$ in $\mathbb{Q}$, then $x<y$ in $\mathbb{R}$.)

Q: Are $\mathbb{N}$, $\mathbb{Z}$ ordered fields?